

ViTextVQA: A large-scale visual question answering dataset and a novel multimodal feature fusion method for Vietnamese text comprehension in images

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ABSTRACT

Visual Question Answering (VQA) is a challenging task that requires the joint understanding of natural language and visual content. While early research primarily focused on recognizing objects and scene context, it often overlooked scene text—an essential source of explicit semantic information. This paper introduces **ViTextVQA** (Vietnamese Text-based Visual Question Answering), the first large-scale Vietnamese dataset specializing in text-based VQA. The dataset contains **over 16,000** images and **over 50,000** question-answer pairs. To tackle this task efficiently, **ViTextBLIP-2** (Vietnamese Text-based Bootstrapped Language-Image Model via Fine-tuning) is proposed, a novel multimodal feature fusion method designed to optimize Vietnamese text-based VQA. Experiments with state-of-the-art models highlight the importance of token ordering in OCR text for answer generation, leading to significant performance improvements. The ViTextVQA dataset is publicly available for research purposes.¹

1. Introduction

In recent years, Visual Question Answering (VQA) has garnered significant attention from researchers in Computer Vision (CV) and Natural Language Processing (NLP). The appeal of VQA has surged with the advent of powerful models like Flamingo (Alayrac et al., 2022), GPT-4 (Achiam et al., 2023), and Gemini (Team et al., 2023), which integrate advanced image-based question-answering capabilities. This has propelled VQA into a phase of robust global development, particularly in resource-rich languages such as English and Chinese. Numerous VQA datasets, including (Biten et al., 2019; Qi et al., 2022; Singh et al., 2019; Wang et al., 2022a), have been released, accompanied by innovative methods showcasing remarkable performance, driven largely by neural and transformer architectures (Biten et al., 2022; Geigle et al., 2023; Kil et al., 2023), which extend beyond traditional approaches.

In Vietnamese, Tran et al. (2021) introduced ViVQA, the first dataset tailored for this task, comprising 15,000 samples derived semi-

automatically from VQA v2. Despite its pioneering role, ViVQA faced scrutiny for its limitations, as later detailed by Nguyen et al. (2023a). Addressing these issues, the OpenViVQA dataset emerged in 2023, designed for open-ended Vietnamese VQA with 11,199 images and 37,914 manually crafted question-answer (QA) pairs. This dataset marked a shift toward open-ended questions and answers, opening new research avenues. Subsequently, the ViOCRvQA dataset (Pham et al., 2025) scaled up efforts with nearly 30,000 images and over 120,000 QA pairs, emphasizing text-based question answering in images.

These datasets still exhibit limitations. The ViVQA dataset did not mention scene text. The OpenViVQA dataset incorporates it in about half of its samples, the free-form answers complicate objective performance assessment. The ViOCRvQA fully focuses on text-based VQA but without scene text.

To minimize these shortcomings, we introduce the Vietnamese Text-based Visual Question Answering (ViTextVQA) dataset, featuring 16,762 images and 50,342 QA pairs. ViTextVQA prioritizes extracting

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¹ <https://huggingface.co/datasets/minhquan6203/ViTextVQA>

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Table 1
Overview of the former VQA datasets in English.

Dataset	Type	Image Source	Annotation Method
DAQUAR (Malinowski & Fritz, 2014)	Non text	NYU-DepthV2	Human annotation
VQA v1 (Antol et al., 2015)	Non text	MS COCO	Human annotation
VQA v2 (Goyal et al., 2017)	Non text	MS COCO	Human annotation
TextVQA (Singh et al., 2019)	Text	MS COCO	Human annotation
OCR-VQA (Mishra et al., 2019)	Text	Book/movie covers	Semi-automation
ST-VQA (Biten et al., 2019)	Text	Multiple sources	Human annotation
DocVQA (Mathew et al., 2021)	Text	Documents, receipts	Human annotation
VisualMRC (Tanaka et al., 2021)	Text	Internet sources	Human annotation



Fig. 1. Samples extracted from the ViTextVQA dataset.

information from scene text to tackle challenges traditional VQA models face (see Fig. 1). Unlike ViVQA, which relied on semi-automatic question generation, the ViTextVQA dataset emphasizes manually curated and text-centric questions to ensure higher quality. In contrast to OpenViVQA, which included scene text in only about half of its samples, ViTextVQA provides comprehensive coverage of text-based questions. Compared with ViOCR-VQA, which focuses on OCR-based VQA but omits scene text, the dataset integrates both elements, offering a more holistic benchmark. To further advance this task, we propose ViTextBLIP-2 (Vietnamese Text-based Bootstrapped Language-Image Model via Fine-tuning), a BLIP-2-based architecture optimized for Vietnamese text-based VQA. In addition to pre-trained vision and language models with a trainable Q-Former for multimodal feature fusion, ViTextBLIP-2 incorporates OCR outputs and image captions to provide both fine-grained textual details and global contextual cues, as detailed in Section 4.

The main contributions of this study are summarized as follows:

- **A new benchmark for Vietnamese text-based VQA.** We present ViTextVQA, a high-quality and large-scale dataset focusing on scene text understanding in Vietnamese. It contains over 16,000 images and 50,000 question-answer pairs with diverse question types and long-form responses, establishing the first comprehensive resource for this task.
- **A unified and efficient multimodal framework.** We propose ViTextBLIP-2, a novel model that leverages frozen pre-trained components and a trainable Q-Former (initialized from BLIP-2) to enable effective integration of visual, textual, and OCR features. This design substantially improves the efficiency and adaptability of VQA systems in low-resource settings.
- **Comprehensive analysis of text layout and OCR ordering.** Through systematic experiments on ViTextVQA, we demonstrate that proper OCR text ordering (from top-left to bottom-right) significantly enhances model performance. This finding provides new insights into the importance of text layout adaptation for multilingual and low-resource VQA tasks.

This article is structured as follows: Section 2 reviews related studies. Section 3 details the ViTextVQA creation process and analysis. Section 4 introduces the proposed ViTextBLIP-2. Section 5 presents baseline models and outlines experimental design and results. Section 6 delves into

result analysis. Section 7 explores Vietnamese-specific traits. Section 8 presents the study's limitations and directions for future research. Finally, Section 9 concludes the study.

2. Related work

2.1. Visual question answering datasets

2.1.1. Former visual question answering datasets

The development of Visual Question Answering (VQA) has been driven by several large-scale English datasets, as summarized in Table 1, which also informed the design of ViTextVQA. Early benchmarks such as DAQUAR (Malinowski & Fritz, 2014) initiated the Visual Turing Challenge by evaluating models' multimodal reasoning. VQA v1 (Antol et al., 2015) introduced open-ended questions and inspired numerous methods (e.g., MCB, MUTAN, NMN), while VQA v2 (Goyal et al., 2017) addressed dataset bias by pairing similar images with different answers.

Scene text understanding became a key focus with datasets like TextVQA (Singh et al., 2019), which introduced 45,336 OCR-based questions and the LoRRA model. OCR-VQA (Mishra et al., 2019) expanded this to over 1 million QA pairs from 207,572 book covers. ST-VQA (Biten et al., 2019) further improved this area by requiring models to reason over text in natural images.

Beyond natural scenes, DocVQA (Mathew et al., 2021) tackled document understanding with 50,000 QA pairs from invoices, forms, and contracts. VisualMRC (Tanaka et al., 2021) advanced this with over 30,000 abstractive QA pairs from 10,000 diverse document images, requiring more complex language generation and document comprehension.

These datasets collectively highlight the evolution of VQA from general image-question tasks to specialized domains involving OCR and document analysis, guiding the creation of ViTextVQA.

2.1.2. Visual question answering datasets in Vietnamese

To develop ViTextVQA, prior Vietnamese VQA datasets (Table 2) are extended to better capture Vietnamese linguistic and cultural nuances. The ViVQA dataset (Tran et al., 2021) was the first of its kind but lacks natural language fluency and excludes text in images. EVJVQA (Nguyen et al., 2023b) provides 33,790 multilingual question-answer pairs from Vietnamese images, yet it overlooks scene text. OpenViVQA (Nguyen et al., 2023a) introduces open-ended answers with 37,914 pairs but is less effective for classification tasks. ViOCR-VQA (Pham et al., 2025) focuses on OCR-VQA with 120,000 pairs from book covers but lacks image diversity. These limitations underscore the need for a more comprehensive and culturally relevant Vietnamese VQA dataset, which ViTextVQA aims to fulfill by offering several key advantages:

- **Diverse contexts and visual content:** It covers a wide range of real-life domains such as education, healthcare, transportation, culture, and daily life in Vietnam.
- **Integration of scene text and visual cues:** It emphasizes understanding from both textual elements embedded in images and visual scenes, enabling more comprehensive multimodal reasoning.

Table 2
Overview of the former VQA datasets in Vietnamese.

Dataset	Type	Image Source	Annotation Method
VIVQA (Tran et al., 2021)	Non text	MS COCO	Semi-Automation
EJVVQA (Nguyen et al., 2023b)	Non text	Internet sources	Human annotation
OpenViVQA (Nguyen et al., 2023a)	Text & Non text	Internet sources	Human annotation
ViOCRQA (Pham et al., 2025)	Text	Internet sources - book covers	Semi-Automation
ViTextVQA (ours)	Text	Internet sources, handcrafted	Human annotation

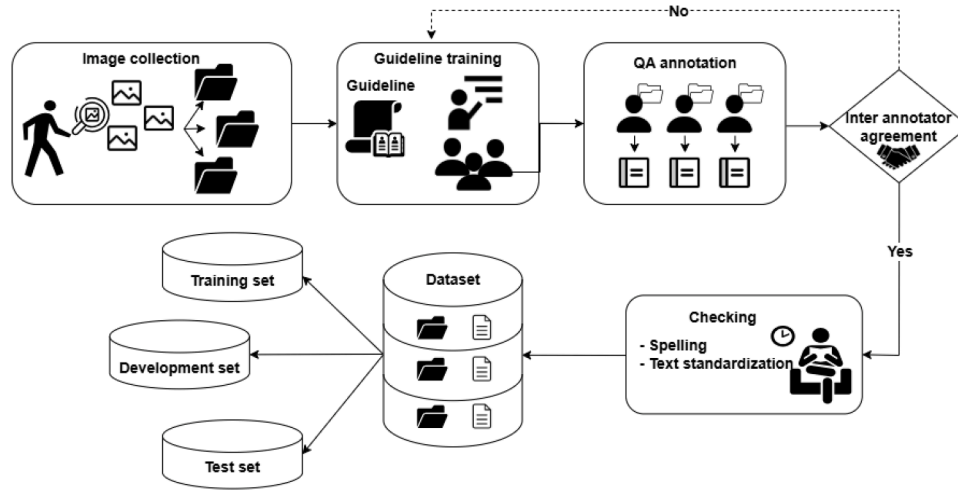


Fig. 2. Overview of the process for creating the ViTextVQA dataset.

- **Natural and semantically rich QA pairs:** Questions and answers are curated with fluent Vietnamese language, making them closer to real-world use cases.
- **Cultural and linguistic relevance:** The dataset reflects Vietnamese-specific vocabulary, expressions, and pragmatic usage, facilitating deeper comprehension of local language and context.

2.2. Visual question answering methods

In this study, VQA models are classified into three main groups: CNN-RNN, CNN-LM, and ViT-LM, based on prior research works (Hu et al., 2024; Rafiepour & Sartakhti, 2023; Wang et al., 2016).

2.2.1. CNN-RNN based methods

CNN-RNN is one of the earliest VQA approaches, combining CNNs for image feature extraction and RNNs for language understanding. Representative models include VIS+LSTM (Strobel et al., 2017) using VGGNet (Simonyan & Zisserman, 2015) and LSTM (Hochreiter & Schmidhuber, 1997), SMem-VQA (Xu & Saenko, 2016) with GoogLeNet (Szegedy et al., 2015) and BoW (Zhang et al., 2010), and Up-Down (Anderson et al., 2018) with ResNet (He et al., 2016) and GRU (Chung et al., 2014). These models integrate visual and textual modalities effectively but struggle with long or complex questions. Specifically, the Up-Down model introduced a bottom-up attention mechanism, where object-level features from Faster R-CNN are dynamically attended to by a top-down RNN controller, enabling more fine-grained reasoning compared to global CNN features.

2.2.2. CNN-LM based methods

Replacing RNNs with Transformer-based language models (e.g., BERT (Devlin et al., 2019)), CNN-LM methods enhance contextual understanding. Notable examples include ViLBERT (Lu et al., 2019), VisualBERT (Li et al., 2020b), Unicoder-VL (Li et al., 2020a), LXMERT (Tan & Bansal, 2019), VL-BERT (Su et al., 2019), UNITER (Chen et al., 2020), and OSCAR (Li et al., 2020c), all of which use CNNs (e.g.,

Faster R-CNN (Girshick et al., 2014)) and transformer LMs with different fusion techniques to improve performance. Among them, LXMERT adopts a dual-stream architecture, where visual and textual features are first encoded separately and then fused through cross-modal attention layers. UNITER, on the other hand, uses a single-stream transformer that jointly encodes concatenated image-region and text embeddings, achieving stronger alignment across modalities.

2.2.3. ViT-LM based methods

Transformers have also advanced computer vision with ViT (Dosovitskiy et al., 2021), leading to ViT-LM models that unify visual and textual transformers. Prominent models include BLIP (Li et al., 2022b), BLIP-2 (Li et al., 2023), GIT (Wang et al., 2022b), mPlug (Li et al., 2022a), Flamingo (Alayrac et al., 2022), BeiT-3 (Wang et al., 2023), and PaLI (Chen & Wang, 2022). These models achieve state-of-the-art results but require extensive computation and training time. For instance, BLIP-2 introduces a query transformer to connect frozen image encoders with large language models, enabling efficient pre-training without fine-tuning the LLM backbone. Flamingo, developed by DeepMind, integrates a visual encoder with a frozen LLM through gated cross-attention layers, allowing the model to handle multi-image and multi-modal in-context learning scenarios.

For further details on resource usage of CNN-LM and ViT-LM models on the ViTextVQA dataset, refer to Section 5.

3. Dataset

3.1. Dataset creation

The creation process of the ViTextVQA dataset is illustrated in Fig. 2. More than 40,000 images were collected from various open sources such as Google,¹ Facebook,² Pinterest,³ and through manual photogra-

¹ <https://www.google.com/>

² <https://www.facebook.com/>

³ <https://www.pinterest.com/>



Fig. 3. Examples of images and QA pairs from the ViTextVQA dataset.

phy. To ensure text relevance, the images were filtered by using Swin-TextSpotter (Huang et al., 2022), retaining over 22,000 high-quality Vietnamese text-rich images. Subsequently, around 16,000 carefully selected images were manually annotated, resulting in approximately 50,000 question-answer (QA) pairs that form the final ViTextVQA dataset.

3.1.1. Image collection

A keyword list related to major Vietnamese cities (e.g., “bảng hiệu ở Thành phố Hồ Chí Minh”, “khu du lịch ở Hội An”) was built to crawl images using Selenium⁴ from platforms like Google Images and Pinterest. Manual collection and local photography were also used to enhance diversity. After filtering, the images were organized into folders for annotation.

3.1.2. Question-answer annotation

Detailed annotation guidelines were defined to ensure that each question is based on visible text in the image, and answers are text spans within the image. Key rules include: no yes/no, location, color, or counting questions; only lowercase answers; and one image having up to 10 QA pairs. Annotators were required to follow consistency rules (e.g., format, spelling) and avoid ambiguous or repetitive content.

3.1.3. Annotation process

A total of 16 trained undergraduate annotators were recruited. Each annotator created at least 600 QA pairs per subset. A team member supervised every 4 annotators to ensure quality. Annotators were trained on 100 sample images and reviewed periodically. Quality control included spelling checks, format normalization, and supervisor validation. After six weeks, over 50,000 question-answer pairs were created. The dataset was randomly split into training, development, and test sets using a 7:1:2 ratio. Fig. 3 illustrates examples of images with their corresponding question-answer pairs from the dataset.

3.1.4. Inter-annotator agreement

To ensure annotation consistency, Inter-Annotator Agreement (IAA) was measured biweekly using an F1-score-based method (Rajpurkar et al., 2016), which is suitable for open-ended QA without fixed label sets. Annotators' answers were compared to reference answers provided by supervisors on 100 samples. The F1-score is calculated based on overlapping words between predictions and ground truth, as detailed in Section 5.4.2. Annotators were required to achieve an F1 score above 70%.

After measuring the Inter-Annotator Agreement through the F1-score twice, the results are summarized in Table 3. In both evaluations, the lowest F1-score values are relatively high (73.91 and 83.36), indicating a considerable level of uniformity between annotators. In addition, the average F1-score increased from the first to the second evaluation, from 81.83 to 85.34, reflecting an improvement in consistency among annotators. These results suggest that the annotated dataset has improved in

Table 3

Results of two rounds of inter-annotator agreement measurements.

	First Time	Second Time
min F1-score (%)	73.91	83.36
max F1-score (%)	87.73	87.05
avg F1-score (%)	81.83	85.34

Table 4

Comparison between well-known VQA datasets.

Dataset	Language	Images	Questions	Answers
TextVQA (Singh et al., 2019)	English	28,408	45,336	453,360
ST-VQA (Biten et al., 2019)		23,038	31,791	31,791
DocVQA (Mathew et al., 2021)		12,767	50,000	50,000
VisualMRC (Tanaka et al., 2021)		10,197	30,562	30,562
VAQA (Kamel et al., 2023)	Arabic	5000	137,888	137,888
BVQA (Bhuyan et al., 2025)	Bangla	3545	17,725	17,725
FM-IQA (Gao et al., 2015)	Chinese	158,392	316,193	316,193
ViVQA (Tran et al., 2021)	Vietnamese	10,328	15,000	15,000
OpenViVQA (Nguyen et al., 2023a)		11,199	21,271	21,271
ViCLEVR (Tran et al., 2024)		26,000	30,000	30,000
ViTextVQA (Ours)		16,762	50,342	50,342

quality over sessions, and the measurement can be considered reliable for confirming the overall dataset quality.

3.2. Dataset analysis

The ViTextVQA dataset includes 16,762 images accompanied by 50,342 QA pairs. Fig. 4 depicts the size of training, development and test sets in the dataset. This dataset offers a diverse range of visual scenes as well as scene texts and corresponding questions and answers. Each image is annotated with some question-answer pairs, providing valuable insights into scene text and object required for accurate model predictions where the answer cannot be outside OCR text in the image and this makes the dataset more special in the VQA task. Moreover, statistical comparisons with other widely used datasets in this field are provided in Table 4.

3.2.1. Question length

The question length statistics are shown in Fig. 5. Question lengths in the dataset vary significantly, ranging from very short (3 tokens) to quite long (31 tokens). This diversity reflects a wide spectrum of question complexity. Short questions typically target basic information, while longer ones often involve detailed or multi-faceted content. The average question length is 9.59 tokens, suggesting that most questions are concise yet informative, balancing clarity with sufficient context.

3.2.2. Answer length

Fig. 6 illustrates the variation in answer lengths, ranging from very short (1 token) to as long as 52 tokens. Despite this wide range, the average answer length is only 4.18 tokens, indicating that most answers are concise. This is primarily due to the annotation constraint that answers must consist solely of text present in the image, resulting in a high frequency of answers from 1 and to tokens.

3.2.3. Object in image

Using VinVL (Zhang et al., 2021), detailed object information such as names, locations, and attributes was extracted, enhancing visual understanding in the ViTextVQA dataset.

Fig. 7 presents the frequency distribution of the top 50 object categories. “Person” is the most frequent object, appearing over 47,000

⁴ <https://www.selenium.dev/>

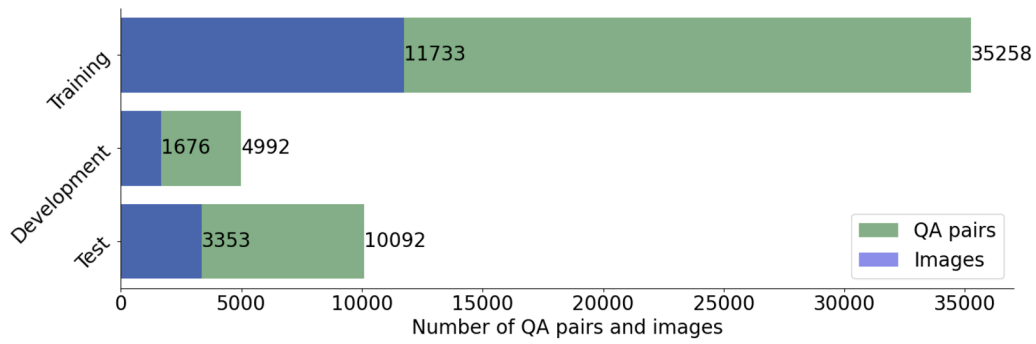


Fig. 4. Statistics of images and question-answer pairs.

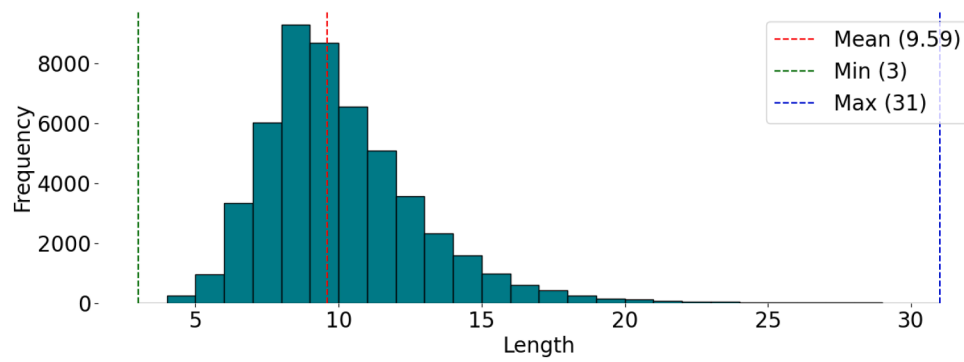


Fig. 5. Distribution of question length.

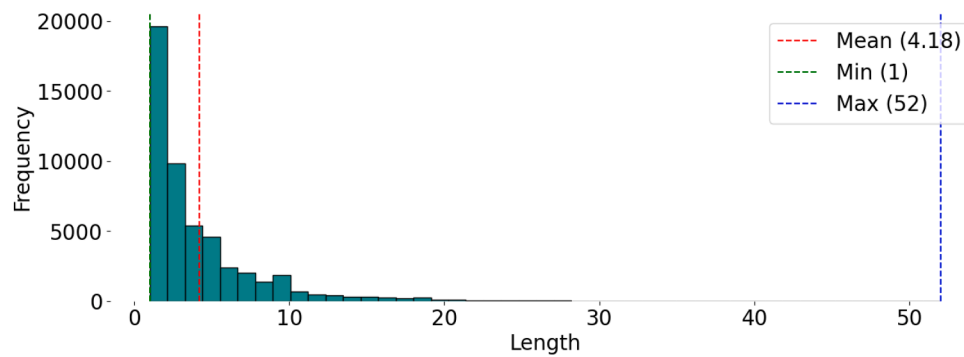


Fig. 6. Distribution of answer length.

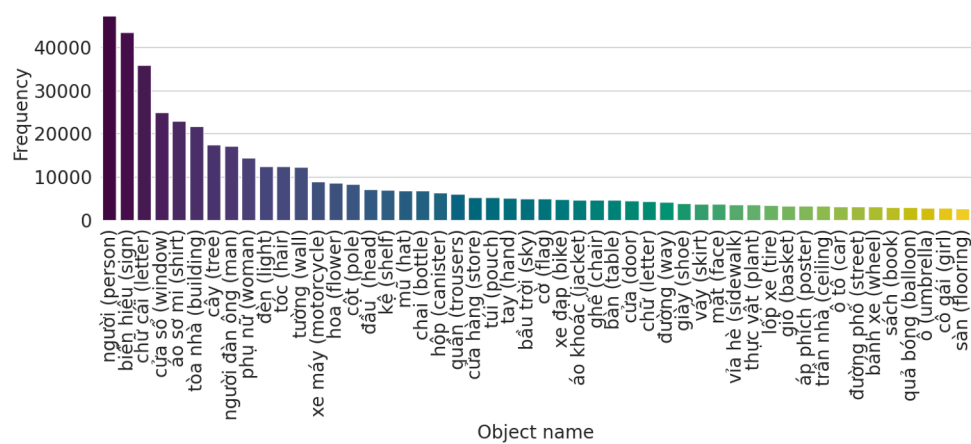


Fig. 7. Distribution of the 50 most frequent objects in images.

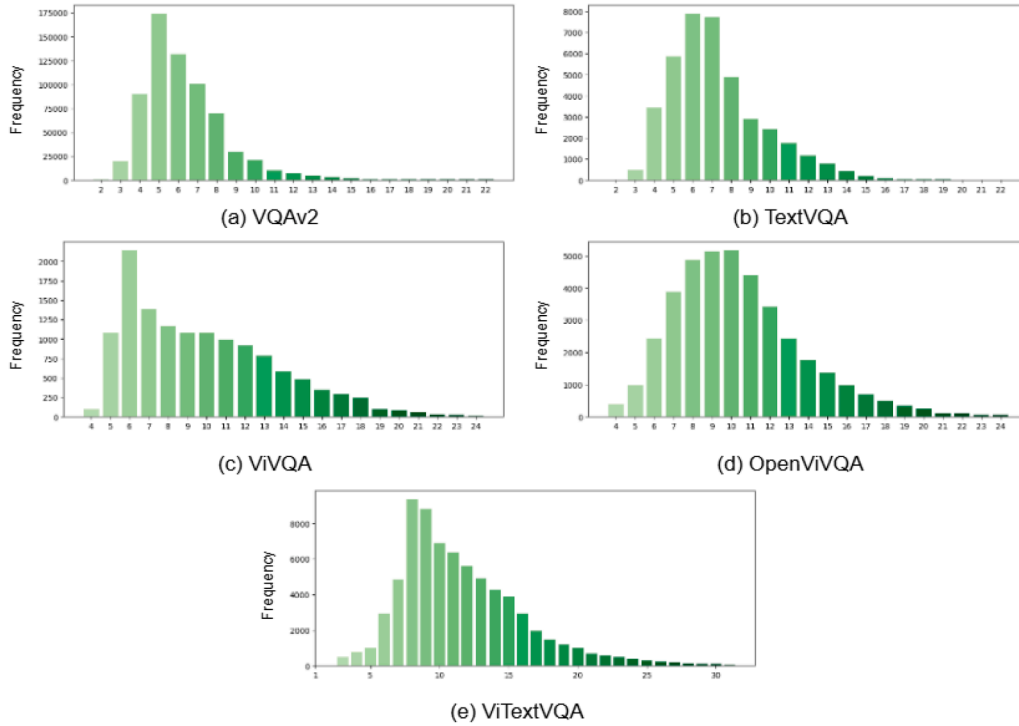


Fig. 8. Comparison of question length among VQA datasets.

times, highlighting the central role of humans in Vietnamese daily life scenes. “Sign” and “letter” follow with 43,000 and 36,000 instances, respectively, reflecting the dataset’s emphasis on scene text and signage-key elements in text-based VQA.

Beyond textual elements, the dataset includes diverse visual contexts, with frequent objects such as “building”, “tree”, “street”, and “sidewalk”. Clothing-related categories like “shirt”, “hat”, and “jacket” also appear frequently, indicating relevance to fashion and lifestyle.

Interestingly, “motorbike” appears nearly 9000 times—about three times more than “car” (over 3000 instances), mirroring Vietnam’s urban mobility patterns and cultural landscape. This object distribution supports the dataset’s contextual richness and cultural alignment.

3.3. Comparison with other visual question answering datasets

To better understand the characteristics of ViTextVQA, comparisons are made with several popular VQA datasets based on two key textual features: question length and answer length. These factors directly affect model performance. Short questions are often easier to process, while longer ones provide richer context and require deeper reasoning. Likewise, longer answers allow for more informative and expressive responses.

As shown in Fig. 8, ViTextVQA contains a higher proportion of long questions compared to VQAv2, TextVQA, and ViVQA, which mainly include short queries. ViTextVQA consistently features questions longer than 15 tokens, making it suitable for tasks that demand detailed understanding. While OpenViVQA also offers diverse question lengths, ViTextVQA includes more high-length examples, increasing its potential for complex reasoning.

Fig. 9 illustrates that ViTextVQA supports longer answers than VQAv2, ViVQA, and TextVQA. Although OpenViVQA shows a more balanced distribution, ViTextVQA allows responses up to 50 tokens, which enables detailed and comprehensive answers. These properties make ViTextVQA a valuable resource for research in visual question answering that involves both visual understanding and text extraction.

4. Methodology

Text-based VQA systems typically consist of three core components: visual feature extraction, textual feature encoding, and an OCR module (Fig. 10). The visual encoder extracts image features, the textual encoder processes the input question, and the OCR module recognizes embedded scene text. The combination of these components enables the system to reason over both visual and textual modalities for answering questions effectively.

ViTextBLIP-2 architecture

ViTextBLIP-2 (Vietnamese Text-based Bootstrapped Language-Image Model via Fine-tuning) (Fig. 11) is proposed as an adaptation of the BLIP-2 framework, optimized for Vietnamese text-based VQA under limited computational and data constraints. Instead of training the entire model, only the Q-Former is fine-tuned while other modules remain frozen, thereby preserving the pre-trained knowledge of large vision and language models and reducing the risk of overfitting on relatively small Vietnamese datasets. By integrating OCR embeddings and captioning signals, the architecture is particularly effective for text-rich images, enabling robust multimodal reasoning. Concentrating the learning process on the Q-Former further enhances cross-modal alignment between visual, textual, and scene-text features.

The inclusion of OCR and caption information strengthens robustness in real-world scenarios, where text-heavy images are prevalent. Moreover, the modular and lightweight design not only reduces training cost but also facilitates adaptability when combining different image encoders or language models without retraining the entire architecture.

Image encoder: A pre-trained Vision Transformer (ViT) is adopted to extract rich visual embeddings from the input image:

$$E_{\text{img}} = \text{ViT}(I), \quad (1)$$

where E_{img} represents image features and I is the input image. Using ViT allows global contextual reasoning across the image and avoids

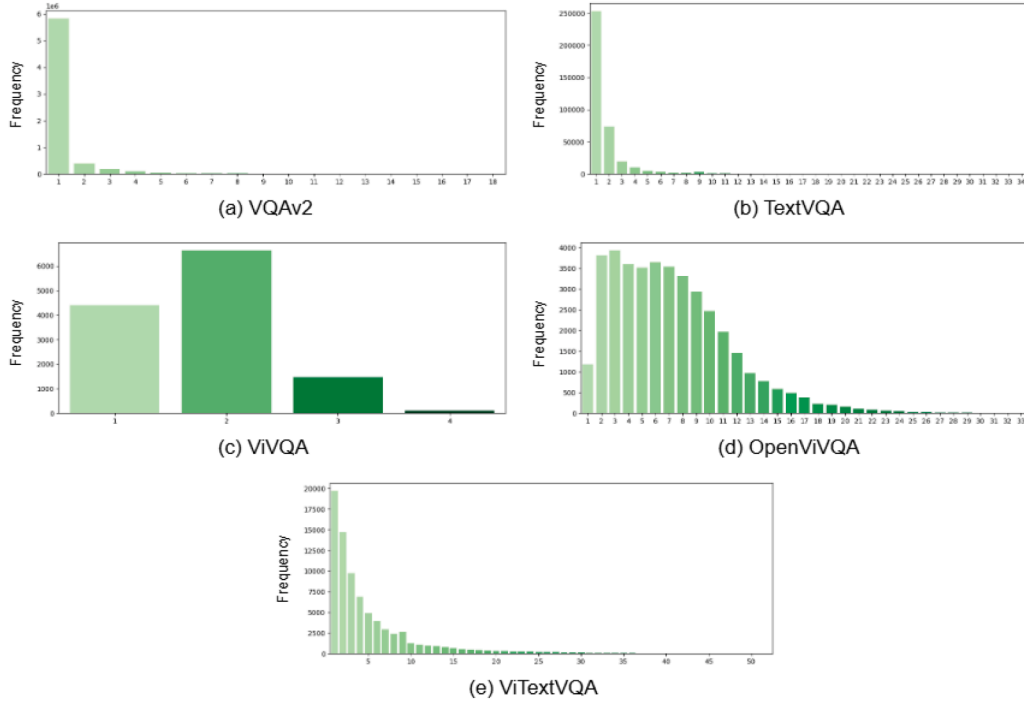


Fig. 9. Comparison of answer length among VQA datasets.

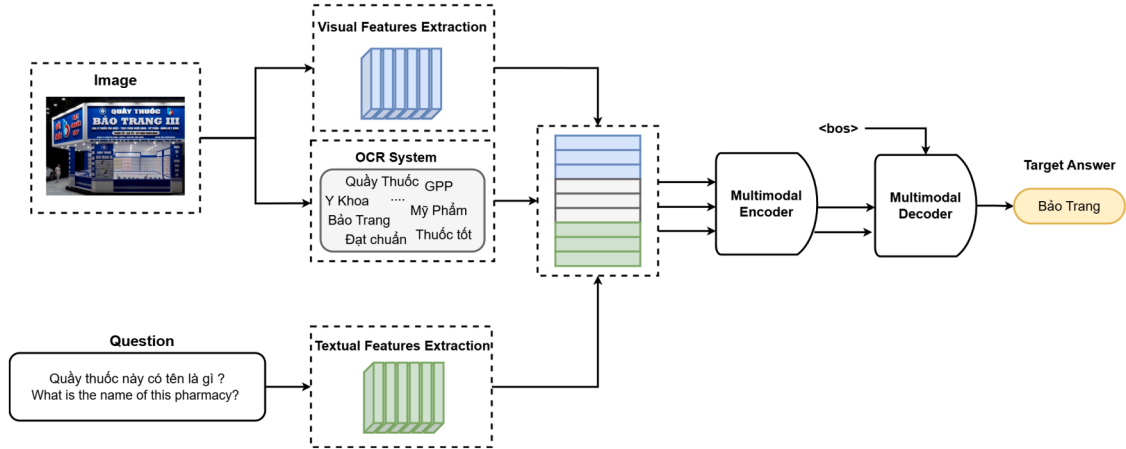


Fig. 10. Overview of the structure of text-based VQA models.

locality limitations of CNNs, which is critical for complex scene understanding in VQA tasks.

OCR system: To handle scene text, SwinTextSpotter (Huang et al., 2022) is integrated to detect and encode text tokens:

$$\{T_{\text{OCR}}, E_{\text{OCR}}\} = \text{SwinTextSpotter}(I), \quad (2)$$

where T_{OCR} are recognized tokens and E_{OCR} are their embeddings. Including OCR embeddings enables the model to answer questions involving text within images, which is common in Vietnamese VQA datasets with signage, menus, or product labels.

Captioning module: To provide semantic context, Qwen2-VL is employed to generate descriptive captions:

$$C = \text{Qwen2VL}(I), \quad (3)$$

where C is the generated caption. Captions help the model by summarizing complex visual scenes into textual form, supporting the LLM in understanding both global context and fine-grained details.

Q-former: The pre-trained Q-Former, initialized from the original BLIP-2 weights, fuses embeddings from the image encoder, OCR system,

and captioning model to support rich contextual reasoning:

$$Q = \text{Q-Former}(E_{\text{img}}, E_{\text{OCR}}, Q_{\text{init}}), \quad (4)$$

where Q is the multimodal representation. Q-Former leverages self-attention to integrate information within each modality and cross-attention to connect visual, textual, and OCR features. This design enables rich multimodal reasoning without updating the entire LLM, reducing computational overhead while maintaining performance.

Large language model: ViT5 (Phan et al., 2022), a Vietnamese pre-trained language model, generates the final answer:

$$A = \text{ViT5}(Q, Q_{\text{ques}}, T_{\text{OCR}}, C), \quad (5)$$

where Q_{ques} is the question embedding and A is the predicted answer. By providing the LLM with a combination of multimodal embeddings, OCR tokens, and captions, the model can handle complex VQA tasks, including those requiring reasoning over text in images. Fine-tuning only the Q-Former ensures low resource usage while effectively aligning image-text representations with the LLM's knowledge.

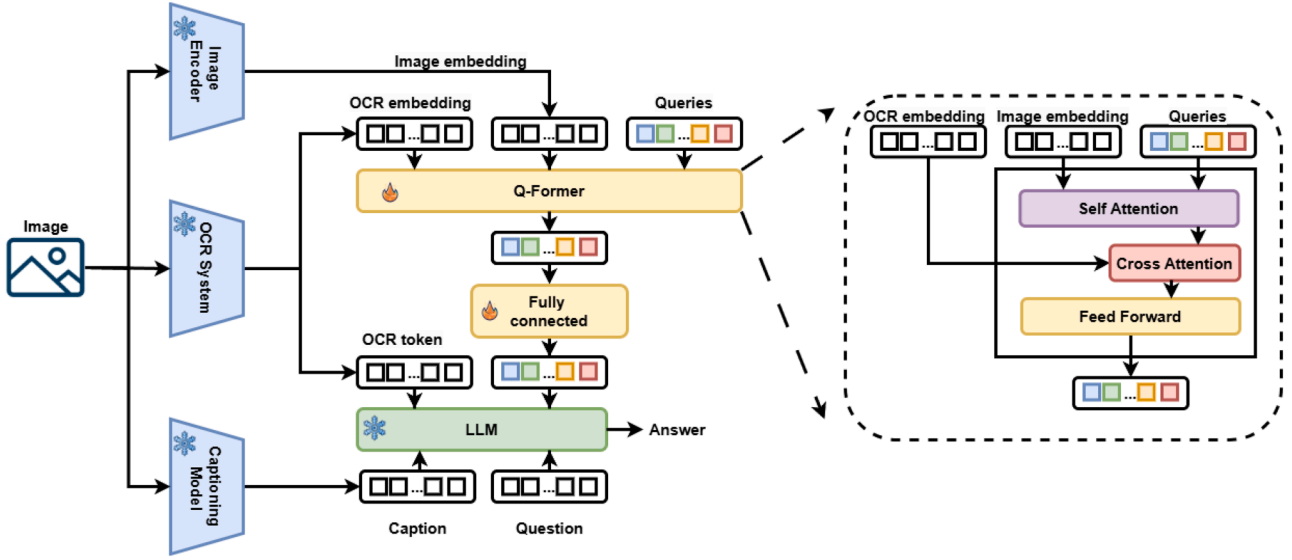


Fig. 11. Architecture of ViTextBLIP-2.

4.1. Feature extraction

To enhance performance, pre-trained models are utilized for textual, visual, and OCR feature extraction. For text encoding, ViT5 (Phan et al., 2022), a Vietnamese encoder-decoder model trained on a large corpus, and mBERT-cased (Pires et al., 2019), a multilingual BERT variant, are adopted. Visual features are extracted using ViT (Dosovitskiy et al., 2021), which divides images into patches for transformer-based processing, and VinVL (Zhang et al., 2021), a Faster R-CNN-based detector trained on diverse objects and attributes. For OCR, SwinTextSpotter (Huang et al., 2022), an open-source Vietnamese-capable text spotter using Swin Transformer for joint detection and recognition, is employed to enable accurate scene text extraction tailored to Vietnamese.

5. Experiment and results

5.1. Baseline models

To assess the difficulty of the ViTextVQA dataset, several state-of-the-art VQA models were evaluated as baselines, selected based on their historical significance and architectural diversity. These include:

M4C (Hu et al., 2020), which introduced a generative VQA framework leveraging BERT-based question embeddings and a Dynamic Pointer Network to integrate OCR and vocabulary tokens;

LaTr (Biten et al., 2022), an encoder-decoder Transformer based on T5, incorporating layout-aware modules for document understanding, including spatial embeddings and visual feature extraction;

PreSTU (Kil et al., 2023), which reorders OCR tokens following their positional layout and applies a T5 architecture pre-trained on scene text image data;

BLIP-2 (Li et al., 2023), which bridges frozen image encoders and large language models using a query transformer for efficient vision-language learning;

SaL (Fang et al., 2023), which addresses the semantic and spatial structure of OCR text via its Text Semantic Separation and Spatial Circle Position modules.

While these models were originally developed for English-language datasets, they were adapted for Vietnamese while retaining their original architectural designs. Table 5 summarizes their key components, including OCR systems, feature extractors, and pre-training strategies, with further implementation details provided in the corresponding sections.

Table 5

Features extraction and OCR system of baseline models.

Model	Visual Features	Textual Features	OCR System	Method Type
M4C	VinVL	mBERT-cased	SwinTextSpotter	CNN-LM
BLIP-2	ViT	ViT5	SwinTextSpotter	ViT-LM
LaTr	ViT	ViT5	SwinTextSpotter	ViT-LM
SaL	VinVL	ViT5	SwinTextSpotter	CNN-LM
PreSTU	ViT	ViT5	SwinTextSpotter	ViT-LM

5.2. Large language models (LLMs)

Recently, large language models (LLMs) such as GPT-4o (Achiam et al., 2023), Gemini 1.5 Flash (Team et al., 2023), and Qwen2-VL (Bai et al., 2023) have significantly advanced the VQA task. GPT-4o demonstrates strong performance in object recognition, OCR, and chart interpretation. Gemini 1.5 Flash processes text and images in real time, enabling fast and accurate visual reasoning. Qwen2-VL supports complex multilingual OCR and chart analysis using techniques like Naive Dynamic Resolution.

5.3. Experimental configuration

The baseline models were trained on a local machine equipped with an RTX 4090 GPU (24GB) using the Adam optimizer (Kingma & Ba, 2015). Training was conducted over 10 epochs, taking approximately 6–7 h for base models (7h for ViT-LM, 6h for CNN-LM) and 9–10 h for large models (10h for ViT-LM, 9h for CNN-LM). Hyperparameters included a learning rate of 3×10^{-5} , dropout rate of 0.2, batch size of 16, and weight decay of 1×10^{-4} . All experiments were run locally on Ubuntu 20.04 with Python 3.10, PyTorch 2.0.1, and Transformers 4.34.0.

5.4. Evaluation metrics

5.4.1. Exact match (EM)

The Exact Match metric calculates the percentage of predictions that match the ground truth answers exactly, providing a strict measure of correctness. It is defined as:

$$EM = \frac{N_{\text{exact}}}{N_{\text{total}}}, \quad (6)$$

where N_{exact} is the number of exact matches and N_{total} is the total number of samples.

Table 6
Results of models on the ViTextVQA test set.

Model	Version	Method Type	Model Size	% Param Tuning	F1-score (%)	EM (%)
M4C	base	CNN-LM	≈140M	≈100%	25.46	10.26
BLIP-2	base	ViT-LM	≈450M	≈25%	36.87	13.59
BLIP-2	large	ViT-LM	≈1.4B	≈8%	38.47	15.58
LaTr	base	ViT-LM	≈330M	≈100%	40.41	17.85
LaTr	large	ViT-LM	≈900M	≈100%	42.34	20.94
Sal	base	CNN-LM	≈270M	≈100%	43.49	20.11
Sal	large	CNN-LM	≈780M	≈100%	43.39	21.48
PreSTU	base	ViT-LM	≈315M	≈100%	43.81	20.85
PreSTU	large	ViT-LM	≈860M	≈100%	44.93	22.64
ViTextBLIP-2 (Ours)	base	ViT-LM	≈2.5B	≈5%	53.95	25.48

Table 7
Performance of LLMs and humans on 100 randomly sampled ViTextVQA test samples.

Model	F1-score (%)				EM (%)			
	0-shot	1-shot	3-shot	5-shot	0-shot	1-shot	3-shot	5-shot
GPT-4o	55.51	69.55	76.67	75.62	25.00	45.00	56.00	55.00
Gemini-1.5-flash	57.75	59.09	62.37	64.40	26.00	29.00	35.00	39.00
QwenVL-7b	53.80	54.10	54.85	54.74	28.00	36.00	36.00	35.00
ViTextBLIP-2	56.43				26.00			
Human	96.42				88.00			

5.4.2. F1-score

F1-score evaluates the overlap between predicted and reference answers at the token level, balancing both precision and recall:

$$\text{Precision} = \frac{N_{\text{correct}}}{N_{\text{predicted}}}, \quad \text{Recall} = \frac{N_{\text{correct}}}{N_{\text{gold}}} \quad (7)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (8)$$

where N_{correct} is the number of overlapping tokens, $N_{\text{predicted}}$ is the number of tokens in the prediction, and N_{gold} is the number of tokens in the reference answer.

5.5. Main results

Table 6 presents the performance of various models evaluated on the ViTextVQA test set using F1-score and EM metrics. Among the baseline methods, the large version of PreSTU (ViT-LM) achieves the best results with an F1-score of 44.93% and an EM of 22.64%. Other competitive models include SaL and LaTr in both base and large configurations, showing that combining visual and textual features with spatial awareness can enhance performance. Overall, the ViT-LM-based approaches generally outperform CNN-LM-based ones, especially in larger versions with higher model capacities (up to ≈900M parameters) and full parameter tuning.

Remarkably, the proposed ViTextBLIP-2 model outperforms all baselines by a significant margin, reaching an F1-score of 53.95% and an EM of 25.48%. Despite having a larger overall size (≈2.5B parameters), ViTextBLIP-2 only fine-tunes a small portion of its parameters (≈5%), specifically the Q-Former module. This lightweight tuning strategy, combined with its effective multimodal design, enables the model to capture both visual semantics and textual information in Vietnamese more efficiently. The results highlight the potential of ViTextBLIP-2 as a strong foundation for advancing VQA research in low-resource language settings.

5.6. Performance of large language models compared to human

In this experiment, LLMs were evaluated on a random subset of 100 samples using 0-shot and few-shot (1, 3, and 5-shot) settings, and the results were compared with human performance.

As shown in Table 7, while GPT-4o and Gemini show performance gains with more shots, all models fall short of human-level accuracy.

Table 8
Experimental results on OpenViVQA test set, inherited from the study of (Nguyen et al., 2023a).

Model	Version	Method Type	CIDEr	Avg BLEU
MCAN	base	ViT-LM	1.0613	0.1699
LoRRA	base	CNN-RNN	0.8005	0.1349
M4C	base	CNN-LM	1.5073	0.2941
FST	base	CNN-LM	0.6141	0.1050
QuMLPAG	base	CNN-LM	1.7082	0.2651
MLPAG	base	CNN-LM	1.6104	0.2739
ViTextBLIP-2 (Ours)	base	ViT-LM	3.2129	0.4717

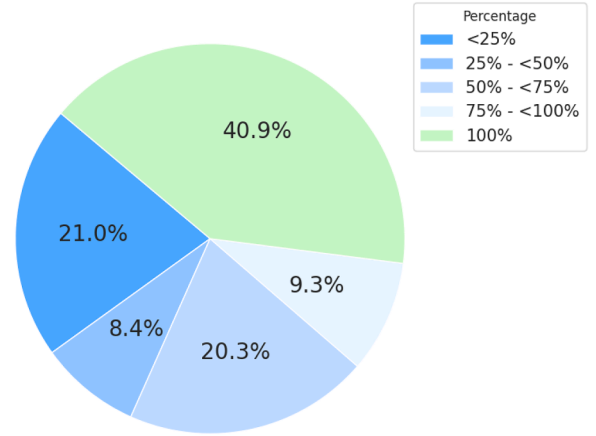


Fig. 12. Proportion of answer tokens found in OCR text on the test set.

Humans achieved 96.42% F1 and 88.00% EM, whereas LLMs peaked at 76.67% F1 and 56.00% EM (GPT-4o, 3-shot). These results highlight the current limitations of LLMs in understanding and reasoning over Vietnamese text in complex visual contexts.

5.7. Compare with other datasets

To assess generalizability beyond ViTextVQA, ViTextBLIP-2 was evaluated on the OpenViVQA dataset (Nguyen et al., 2023a), which features open-ended visual questions. As shown in Table 8, ViTextBLIP-2 achieves a CIDEr score of 3.2129 and an average BLEU score of 0.4717, significantly outperforming both traditional models (e.g., M4C, MCAN, LoRRA) and recently proposed methods (e.g., QuMLPAG, MLPAG). These results demonstrate the effectiveness and robustness of the model across diverse VQA benchmarks.

6. Results analysis

6.1. Effect of OCR system performance and challenges

The performance of VQA models exhibits a strong correlation with OCR quality. Using SwinTextSpotter, test samples were categorized based on the percentage of answer tokens detected in the OCR output. The categories include: <25%, 25–50%, 50–75%, 75–100%, and 100% (see Fig. 12).

As shown in Fig. 13, F1-score generally increases with higher OCR token coverage, confirming that OCR-extracted text plays a vital role in VQA prediction. However, EM results fluctuate, with drops at mid-ranges (25–50% and 75–100%), then rising sharply at full coverage (100%).

Despite this, even with full token coverage, the best F1 and EM remain below 60 and 35, respectively. This reflects the challenge of ViTextVQA: models must not only locate correct text spans but also filter noise and integrate visual-textual context effectively.

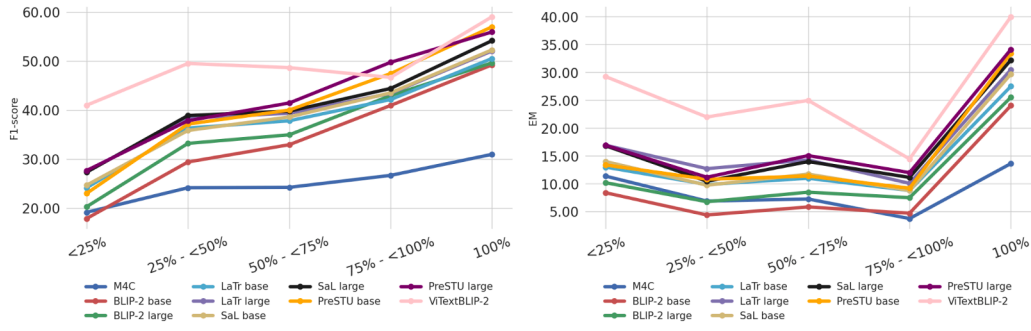


Fig. 13. Model performance (F1 and EM) with respect to OCR token coverage.

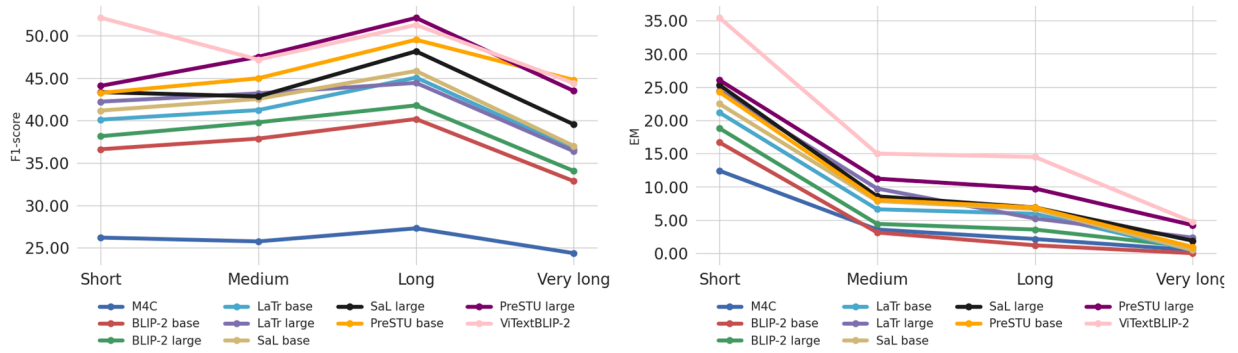


Fig. 14. Model performance by answer length.

Table 9

Distribution of answer and question lengths in the test set.

Group	Length (n)	Answer Samples	Question Samples
Short	$n \leq 5$	7,853	225
Medium	$5 < n \leq 10$	1,541	6,834
Long	$10 < n \leq 15$	421	2,617
Very long	$n > 15$	213	352

6.2. Effect of answer and question length

Samples in the test set were grouped by token length (Table 9) to examine how question and answer length affect model performance. The results reveal clear patterns.

As shown in Fig. 14, model performance tends to decline as answer length increases. EM scores drop steadily with longer answers, reflecting reduced exact match capability. For F1-score, medium and long answers yield better performance than short ones, but this trend reverses with very long answers, likely due to accumulated OCR noise or increased linguistic complexity.

Regarding question length (Fig. 15), shorter questions achieve the highest F1 scores. Performance gradually decreases as length increases, with a sharp drop for very long questions. For EM, the performance remains relatively stable across short to long ranges, though individual model behaviors vary. For instance, SaL base improves with question length up to 15 tokens, while PreSTU base declines. Very long questions pose notable challenges for all models, reducing both EM and F1 scores.

6.3. Effect of OCR text size

To assess the impact of OCR text size on model performance, only samples in which all answer tokens appeared in the OCR output (as

in Section 6.1) were analyzed to ensure reliable measurement. Using VinVL (Zhang et al., 2021), the total bounding-box area of answer tokens was computed as a percentage of the image area. Based on this ratio, samples were grouped as: *small* ($< 1\%$), *medium* ($1\text{--}5\%$), and *large* ($> 5\%$) (Fig. 16).

As shown in Fig. 17, F1-score consistently improves with larger OCR text regions across all models, indicating better recognition and contextual grounding. For EM, results are less consistent: performance drops slightly from small to medium sizes but improves again in the large group. This suggests that while larger text regions generally enhance comprehension, exact matching remains sensitive to OCR segmentation quality.

6.4. Effect of OCR arrangement

In this experiment, the impact of different OCR text arrangements on VQA model performance is investigated. While scene text typically follows a top-left to bottom-right (TL-BR) reading order, OCR outputs may not preserve this structure. Three strategies are therefore evaluated: (1) original order, (2) sorting by OCR confidence (descending), and (3) TL-BR sorting (Kil et al., 2023).

Experiments on both ViTextVQA and OpenViVQA datasets (Tables 10 and 11) reveal consistent trends. Most models benefit from TL-BR sorting, achieving higher F1/EM or CIDEr/BLEU scores. This aligns with Vietnamese reading patterns, enabling models to better preserve textual semantics and sequence.

Conversely, confidence-based sorting generally degrades performance. Rearranging tokens by confidence disrupts the spatial and linguistic coherence of the text, making interpretation more difficult.

Overall, OCR arrangement plays a significant role in model accuracy. Maintaining natural reading order—specifically TL-BR, enhances comprehension, while artificial reordering (e.g., confidence-based) may hinder it.

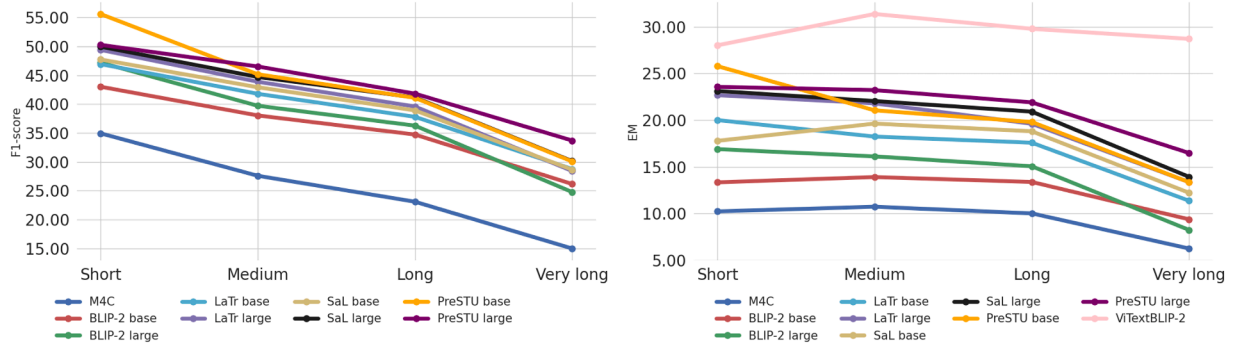


Fig. 15. Model performance by question length.



Fig. 16. Illustration of OCR text size groups.

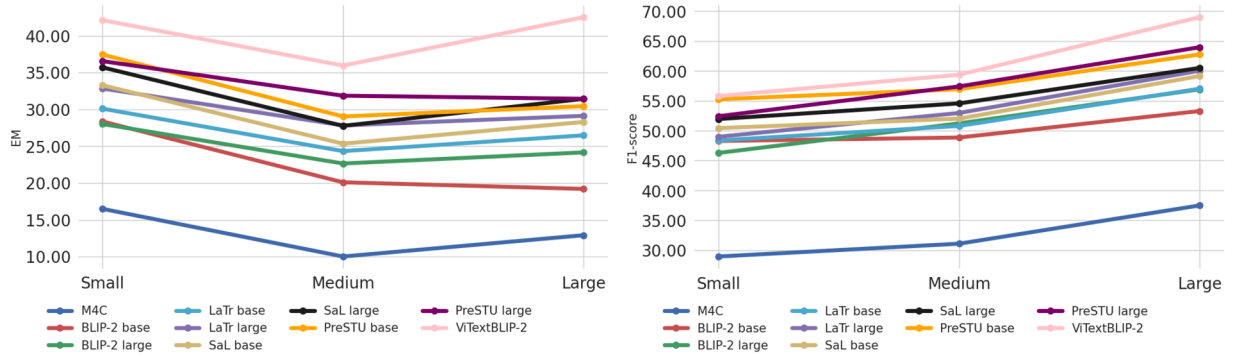


Fig. 17. Model performance by OCR text size.

Table 10

Results of models with different OCR arrangements on ViTextQA dataset. Δ indicates the increase $\uparrow$ or decrease $\downarrow$ compared with the original (*).

Model	Original Order		OCR Confidence		TL-BR	
	F1-score (%)	EM (%)	F1-score (%)	EM (%)	F1-score (%)	EM (%)
M4C	25.46*	10.26*	18.41 ($\downarrow 7.05$)	8.29 ($\downarrow 1.97$)	30.04 ($\uparrow 4.58$)	11.60 ($\uparrow 1.34$)
BLIP-2 base	36.87*	13.59*	35.21 ($\downarrow 1.66$)	12.97 ($\downarrow 0.62$)	37.78 ($\uparrow 0.91$)	15.01 ($\uparrow 1.42$)
BLIP-2 large	38.47*	15.58*	36.74 ($\downarrow 1.73$)	14.15 ($\downarrow 1.43$)	42.16 ($\uparrow 3.69$)	18.56 ($\uparrow 2.98$)
LaTr base	40.41*	17.85*	38.76 ($\downarrow 1.65$)	16.32 ($\downarrow 1.53$)	43.13 ($\downarrow 2.72$)	20.42 ($\downarrow 2.57$)
LaTr large	42.34*	20.94*	39.72 ($\downarrow 2.62$)	18.88 ($\downarrow 2.06$)	44.07 ($\uparrow 1.73$)	22.27 ($\uparrow 1.33$)
SaL base	43.49*	20.11*	43.66 ($\uparrow 0.17$)	20.28 ($\uparrow 0.17$)	44.89 ($\uparrow 1.40$)	20.97 ($\uparrow 0.86$)
SaL large	43.39*	21.48*	43.71 ($\uparrow 0.32$)	20.47 ($\uparrow 0.31$)	44.74 ($\uparrow 1.35$)	21.18 ($\uparrow 0.30$)
PreSTU base	41.74 ($\downarrow 2.07$)	19.15 ($\downarrow 1.70$)	40.09 ($\downarrow 3.72$)	17.74 ($\downarrow 3.11$)	43.81*	20.85*
PreSTU large	42.85 ($\downarrow 2.08$)	21.40 ($\downarrow 1.24$)	40.13 ($\downarrow 4.80$)	18.43 ($\downarrow 4.21$)	44.93*	22.64*
ViTextBLIP-2 (Ours)	51.16 ($\uparrow 1.35$)	24.10 ($\uparrow 0.73$)	53.95 ($\uparrow 1.44$)	25.48 ($\uparrow 0.65$)	52.51*	24.83*
Avg	40.62	18.45	39.04	17.30	42.81	19.83

Table 11

Results of ViTextBLIP-2 with different OCR arrangements on OpenViQA dataset. Δ indicates the increase $\uparrow$ or decrease $\downarrow$ compared with the original (*).

Model	Original Order		OCR Confidence		TL-BR	
	CIDEr	Avg BLEU	CIDEr	Avg BLEU	CIDEr	Avg BLEU
ViTextBLIP-2 (Ours)	2.6329 ($\downarrow 0.58$)	0.4125 ($\downarrow 0.06$)	3.0024 ($\downarrow 0.21$)	0.4502 ($\downarrow 0.02$)	3.2129*	0.4717*

Table 12

Ablation study on different model components.

Image Encoder	OCR System	Captioning Model	F1-score (%)	EM (%)
✓	✓	✓	0.5251*	0.2483*
✓	✓	✗	0.3914 ($\downarrow 0.1337$)	0.1694 ($\downarrow 0.0789$)
✓	✗	✓	0.3953 ($\downarrow 0.1298$)	0.1794 ($\downarrow 0.0689$)
✓	✗	✗	0.1417 ($\downarrow 0.3834$)	0.0469 ($\downarrow 0.2014$)
✗	✓	✓	0.4167 ($\downarrow 0.1084$)	0.1882 ($\downarrow 0.0601$)
✗	✓	✗	0.3668 ($\downarrow 0.1583$)	0.1332 ($\downarrow 0.1511$)
✗	✗	✓	0.3512 ($\downarrow 0.1739$)	0.1267 ($\downarrow 0.1216$)
✗	✗	✗	0.0016 ($\downarrow 0.5235$)	0.0000 ($\downarrow 0.2483$)

6.5. Ablation study

In this section, an ablation study is presented to evaluate the contribution of each component in ViTextBLIP-2: Image Encoder, OCR System, and Captioning Model. The results are summarized in Table 12.

Full model (✓✓✓): The complete model achieves the best performance (F1: 52.51 %, EM: 24.83 %), confirming that the combination of visual, textual, and semantic features is essential for robust performance.

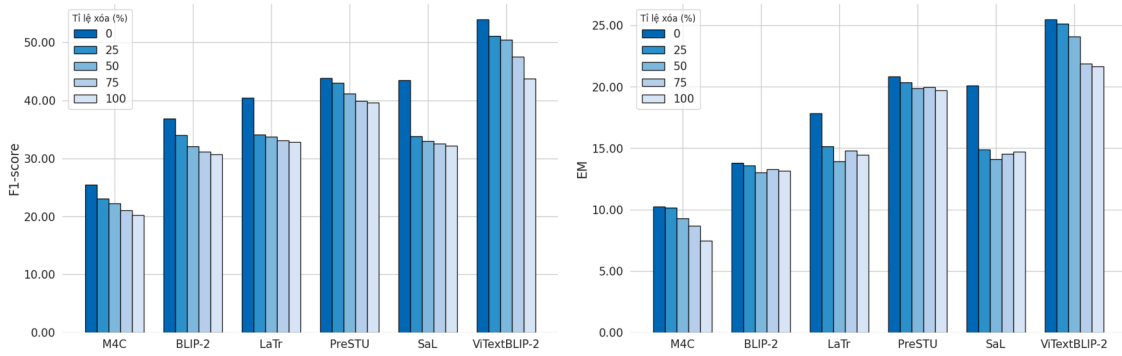


Fig. 18. Performance of base models under different diacritic removal rates.

Removing captioning (✓/✗): Excluding the captioning module leads to a notable drop (F1: ↓13.37 %, EM: ↓7.89 %), indicating its role in providing complementary semantic cues.

Removing OCR (✓/✗): Without OCR, performance degrades similarly (F1: ↓12.98 %, EM: ↓6.89 %), emphasizing the importance of scene text for text-based VQA.

Image encoder only (✓/✗): Using only the Image Encoder results in poor performance (F1: 14.17 %, EM: 4.69 %), showing it alone cannot support complex text-based reasoning.

No components (✗/✗): Removing all modules renders the model ineffective (F1: 0.16 %, EM: 0.00 %), highlighting the necessity of multimodal integration.

Overall, the study confirms that each module contributes significantly to ViTextBLIP-2, with OCR and captioning providing critical text and context to support accurate predictions.

7. Discussion on characteristics of language for Vietnamese text-based visual question answering

For a language such as Vietnamese, unique characteristics must be considered to achieve high performance in the VQA task. This section analyzes and discusses two main aspects of the Vietnamese language: diacritics and word segmentation, as well as their relationship to this task.

7.1. Impact of diacritics on Vietnamese text understanding

Vietnamese is a tonal language with a rich diacritic system comprising seven letters (ã, â, ê, ô, o, u, đ) and five tonal marks (à, á, â, ã, ạ) (Nguyen et al., 2019). These diacritics are essential for distinguishing phonemes and lexical meanings, e.g., “má” (mother) vs. “mà” (but). Historically, their development was shaped by Chinese and Nom influences, and later standardized during the colonial era.

To evaluate the role of diacritics, experiments were conducted on the ViTextVQA dataset by progressively removing diacritics at rates of 0 %, 25 %, 50 %, 75 %, and 100 %.

As shown in Fig. 18, removing diacritics leads to a significant performance drop across all models. F1-score decreases consistently with higher removal rates, confirming that diacritics are critical for semantic understanding. EM scores fluctuate more but generally decline, with the sharpest drop at 50 % removal for most models.

Notably, LaTr and SaL are more sensitive to diacritic loss, especially from 0 % to 25 %, suggesting a strong reliance on diacritic information. These findings highlight the importance of diacritics in preserving meaning and improving model comprehension in Vietnamese VQA tasks.

7.2. Impact of Vietnamese word segmentation

Vietnamese presents unique challenges in word segmentation due to the lack of explicit word boundaries. Unlike English, white spaces

Table 13

Word segmentation statistics under different OCR arrangements.

Arrangement Method	Min	Max	Avg	Avg EM (%)	Avg F1-score (%)
Original Order	0	11	0.56	18.45	40.62
OCR Confidence	0	10	0.56	17.30	39.04
TL-BR	0	14	1.06	19.83	42.81

in Vietnamese separate syllables, not necessarily words (Nguyen et al., 2018). For example, “thời gian” (time) consists of two syllables forming a single word “thời gian”. Accurate segmentation is therefore essential for correct semantic interpretation.

We hypothesize that OCR text arrangements which better preserve the natural reading order of Vietnamese can lead to improved word segmentation and thus better performance. Using VnCoreNLP (Vu et al., 2018), segmented word counts were analyzed across three OCR arrangements.

As shown in Table 13, the top-left to bottom-right (TL-BR) strategy yields the highest average number of segmented words and achieves the best model performance. This ordering aligns with natural Vietnamese reading patterns, allowing better context reconstruction and more accurate segmentation.

These results suggest that optimizing OCR arrangements to increase word segmentation quality can be an effective strategy for enhancing performance in Vietnamese VQA and other NLP tasks where word-level understanding is critical.

8. Limitations and future work

While the ViTextVQA dataset provides a resource for Vietnamese multimodal research, several limitations remain. First, the performance of current models is highly sensitive to OCR quality, which introduces errors that propagate into downstream predictions. Second, ViTextVQA primarily focuses on scene text understanding, which may limit its applicability in scenarios where visual reasoning extends beyond textual elements.

Additionally, the ViTextBLIP-2 model was trained specifically for Vietnamese and has not yet been evaluated in multilingual or zero-shot settings. Exploring cross-lingual transfer, robust OCR techniques, and visual reasoning beyond text-centric questions represents valuable directions for future research.

Looking forward, notable research directions can be highlighted, including:

- Adapt the dataset for Visual Question Generation (VQG), where the task involves generating contextually relevant questions given an image and an answer (Fan et al., 2018; Xu et al., 2020; Zhang et al., 2017). Beyond conventional VQA benchmarks, future work can extend these models to domains such as environmental and climate-related applications, which require advanced reasoning over

multimodal information (Guo et al., 2023, 2024a,b). Considering recent findings on systematic poisoning attacks in healthcare ML systems (Mozaffari-Kermani et al., 2014), security-conscious data adaptation will also be essential to ensure robustness against adversarial inputs.

- Investigate prompting-based methods with large language models (LLMs), inspired by promising results from recent studies (Jia et al., 2022; Zhou et al., 2022), which demonstrate strong performance even under limited fine-tuning. A natural extension is the development of a Vietnamese multimodal chatbot - similar in spirit to Flamingo (Alayrac et al., 2022), GPT-4 (Achiam et al., 2023), and Gemini (Team et al., 2023) - capable of handling image-grounded queries in Vietnamese, thereby advancing localized multimodal understanding and interaction.

9. Conclusion

In this work, we introduced ViTextVQA, a large-scale Vietnamese Visual Question Answering dataset, comprising over 16,000 images and 50,000 question-answer pairs. The dataset is designed to advance research in Vietnamese multimodal understanding, addressing the underrepresentation of low-resource languages in VQA. Unlike ViVQA, which relied on semi-automatic question generation, OpenViVQA, which only partially incorporated scene text, or ViOCRvQA, which omitted it entirely. We hope ViTextVQA will contribute meaningfully to the diversity of available linguistic resources and foster further research in Vietnamese language technologies.

We also benchmarked several state-of-the-art VQA models on ViTextVQA and observed that their performance, while strong on English datasets, does not translate equivalently to Vietnamese. This gap motivated a thorough performance analysis, uncovering key insights-most notably, that arranging OCR text in a top-left to bottom-right order significantly enhances model accuracy. Compared with baseline models such as BLIP-2, LaTr, and PreSTU, the proposed method ViTextBLIP-2 achieves substantial improvements, outperforming them by a margin of 53.95 % in overall F1-score on ViTextVQA. These findings highlight the need to tailor preprocessing and modeling strategies to the linguistic and structural characteristics of Vietnamese.

CRedit authorship contribution statement

Quan Van Nguyen: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft; **Dan Quang Tran:** Conceptualization, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft; **Huy Quang Pham:** Conceptualization, Data curation, Investigation, Methodology, Writing – original draft; **Thang Kien-Bao Nguyen:** Conceptualization, Data curation, Investigation, Methodology, Writing – original draft; **Nghia Hieu Nguyen:** Conceptualization, Data curation, Investigation, Methodology, Writing – review & editing; **Kiet Van Nguyen:** Conceptualization, Formal analysis, Investigation, Methodology, Validation, Supervision, Writing – review & editing; **Ngan Luu-Thuy Nguyen:** Conceptualization, Formal analysis, Investigation, Methodology, Validation, Supervision, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Data creation rules

General rules

General Rules are established to ensure that the process of creating questions and answers for images is done carefully and systematically. This regulation helps ensure consistency and quality of data, from ensuring the number of questions and answers per image.

- **Rule 1:** For each subset, the annotator created at least 600 pairs of questions and answers.
- **Rule 2:** The annotator is encouraged to use their personal vocabulary to ask questions.
- **Rule 3:** Each image contains at most 10 pairs of questions and answers.
- **Rule 4:** All letters are in lowercase and in the scope of the standard keyboard.

Rules for creating questions

The rules for creating questions provide specific instructions on how to ask questions so that they are specific, not vague, and ensure that the question leads to an answer that is already in the image. This way, not only does it create valuable questions, but it also creates convenience for participants to think and reason creatively and logically.

- **Rule 1:** Question must be answered by text that appears only in the image.
- **Rule 2:** Do not ask questions with mathematical implications.
- **Rule 3:** Do not ask where the text is located.
- **Rule 4:** Do not ask options (yes/no).
- **Rule 5:** Do not ask the color of the text.
- **Rule 6:** Do not ask the number of text that the annotator has to count.
- **Rule 7:** Avoid asking questions that are too short.
- **Rule 8:** Avoid asking multiple questions that have the same answer in the same image.
- **Rule 9:** Avoid asking general questions that lead to many different answers.
- **Rule 10:** Take the direction of the annotator seeing the image as the reference.

Rules for creating answers

The rule for creating answers guides annotators to make sure that annotated answers are correct and do not contain extraneous information. This ensures the consistency of the dataset and makes the models easier to process.

- **Rule 1:** The answer is only the words that appear in the image.
- **Rule 2:** Keep the content intact, convert everything to lowercase.
- **Rule 3:** If the text has a line, replace it with a space.
- **Rule 4:** If the answer has some words that are obscured, only annotate what the annotator sees.

Appendix B. OCR system

SwinTextSpotter: Since OCR quality strongly influences VQA performance, ViTextBLIP-2 incorporates SwinTextSpotter (Huang et al.,

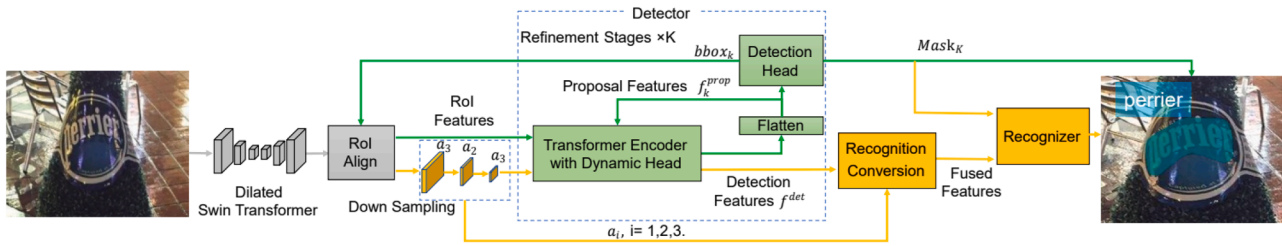


Fig. B.1. The overview of SwinTextSpotter structure.

2022) as the OCR backbone due to its state-of-the-art accuracy and support for multilingual scene text spotting, including Vietnamese. SwinTextSpotter is an end-to-end framework that tightly integrates text detection and recognition using a shared Swin Transformer backbone, enhancing the synergy between these two stages (see Fig. B.1).

In the present pipeline, the publicly available pre-trained SwinTextSpotter model is employed without additional fine-tuning. To enable language-specific customization, the text recognition decoder is configured with Vietnamese character sets and diacritic-aware settings. This adaptation ensures accurate transcription of tone-sensitive Vietnamese scene text, which is critical for downstream VQA tasks that require fine-grained linguistic understanding.

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